



ENVIRONMENTAL DATA ANALYTICS: M4 – DATA WRANGLING

Agenda

- M3-Data Exploration Lab/Assignment Q&A
 - ▣ *Formatting your document for knitting...*
- M4-Data Wrangling Lesson Review
- M4-Data Wrangling Exercises
 - ▣ *Peter & Paul Lakes...*
 - ▣ *Filter | Pipe | Gather & Spread*
- M4-Assignment

M4.1

Q&A on Data Wrangling

- Datasets, “Tidy Data”
- Importing data
- Wrangling data with `dplyr`
 - | `filter` | `arrange` | `select` | `mutate` | ← *covered*
 - | `slice` | `rename` | `relocate` | `summarize` | ← *vignette*

Data transformation with dplyr :: CHEAT SHEET



dplyr functions work with pipes and expect **tidy data**. In tidy data:



Each **variable** is in its own **column**

&



Each **observation**, or **case**, is in its own **row**



pipes

$x \%>\% f(y)$ becomes $f(x, y)$

Summarise Cases

Apply **summary functions** to columns to create a new table of summary statistics. Summary functions take vectors as input and return one value (see back).

summary function



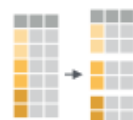
summarise(.data, ...)
Compute table of summaries.
`summarise(mtcars, avg = mean(mpg))`



count(.data, ..., wt = NULL, sort = FALSE, name = NULL) Count number of rows in each group defined by the variables in ... Also **tally()**.
`count(mtcars, cyl)`

Group Cases

Use **group_by(.data, ..., add = FALSE, .drop = TRUE)** to create a "grouped" copy of a table grouped by columns in ... dplyr functions will manipulate each "group" separately and combine the results.



`mtcars %>%
group_by(cyl) %>%
summarise(avg = mean(mpg))`

Use **rowwise(.data, ...)** to group data into individual rows. dplyr functions will compute results for each row. Also apply functions to list-columns. See tidy cheat sheet for list-column workflow.



`starwars %>%
rowwise() %>%
mutate(film_count = length(films))`

ungroup(x, ...) Returns ungrouped copy of table.
`ungroup(g_mtcars)`

Manipulate Cases

EXTRACT CASES

Row functions return a subset of rows as a new table.



filter(.data, ..., .preserve = FALSE) Extract rows that meet logical criteria.
`filter(mtcars, mpg > 20)`



distinct(.data, ..., .keep_all = FALSE) Remove rows with duplicate values.
`distinct(mtcars, gear)`



slice(.data, ..., .preserve = FALSE) Select rows by position.
`slice(mtcars, 10:15)`



slice_sample(.data, ..., n, prop, weight_by = NULL, replace = FALSE) Randomly select rows. Use `n` to select a number of rows and `prop` to select a fraction of rows.
`slice_sample(mtcars, n = 5, replace = TRUE)`



slice_min(.data, order_by, ..., n, prop, with_ties = TRUE) and **slice_max()** Select rows with the lowest and highest values.
`slice_min(mtcars, mpg, prop = 0.25)`



slice_head(.data, ..., n, prop) and **slice_tail()** Select the first or last rows.
`slice_head(mtcars, n = 5)`

Logical and boolean operators to use with filter()

<code>==</code>	<code><</code>	<code><=</code>	<code>is.na()</code>	<code>%in%</code>	<code> </code>	<code>xor()</code>
<code>!=</code>	<code>></code>	<code>>=</code>	<code>is.na()</code>	<code>!</code>	<code>&</code>	

See `?base::Logic` and `?Comparison` for help.

ARRANGE CASES



arrange(.data, ..., .by_group = FALSE) Order rows by values of a column or columns (low to high), use with **desc()** to order from high to low.
`arrange(mtcars, mpg)`
`arrange(mtcars, desc(mpg))`

ADD CASES



add_row(.data, ..., .before = NULL, .after = NULL) Add one or more rows to a table.
`add_row(cars, speed = 1, dist = 1)`

Manipulate Variables

EXTRACT VARIABLES

Column functions return a set of columns as a new vector or table.



pull(.data, var = -1, name = NULL, ...) Extract column values as a vector, by name or index.
`pull(mtcars, wt)`



select(.data, ...) Extract columns as a table.
`select(mtcars, mpg, wt)`



relocate(.data, ..., .before = NULL, .after = NULL) Move columns to new position.
`relocate(mtcars, mpg, cyl, .after = last_col())`

Use these helpers with select() and across()

e.g. `select(mtcars, mpg:cyl)`

contains(match)	num_range(prefix, range)	*, e.g. <code>mpg:cyl</code>
ends_with(match)	all_of(x)/any_of(x, ..., vars)	*, e.g. <code>-gear</code>
starts_with(match)	matches(match)	everything()

MANIPULATE MULTIPLE VARIABLES AT ONCE



across(.cols, .funs, ..., .names = NULL) Summarise or mutate multiple columns in the same way.
`summarise(mtcars, across(everything(), mean))`



c_across(.cols) Compute across columns in row-wise data.
`transmute(rowwise(UKgas), total = sum(c_across(1:2)))`

MAKE NEW VARIABLES

Apply **vectorized functions** to columns. Vectorized functions take vectors as input and return vectors of the same length as output (see back).

vectorized function



mutate(.data, ..., .keep = "all", .before = NULL, .after = NULL) Compute new column(s). Also **add_column()**, **add_count()**, and **add_tally()**.
`mutate(mtcars, gpm = 1 / mpg)`



transmute(.data, ...) Compute new column(s), drop others.
`transmute(mtcars, gpm = 1 / mpg)`



rename(.data, ...) Rename columns. Use **rename_with()** to rename with a function.
`rename(cars, distance = dist)`

dplyr – test your knowledge!

□ <https://PollEv.com/johnfay695>

ENV872-DataWrangling

5 questions

Start survey

Q&A: dplyr

Filter	Arrange	Select	Mutate	Pipe	Lubridate
--------	---------	--------	--------	------	-----------

Perform multiple operations on a data frame...

```
NTL.phys.data.processed <-  
  NTL.phys.data %>%  
  filter(lakename == "Paul Lake" | lakename == "Peter Lake") %>%  
  select(lakename, sampleddate:temperature_C) %>%  
  mutate(temperature_F = (temperature_C*9/5) + 32)
```

lakeid	lakename	year4	daynum	sampledate	depth	temperature_C	dissolvedOxygen	irradianceWater
L	Paul Lake	1984	148	1984-05-27	0.00	14.5	9.5	1750.0
L	Paul Lake	1984	148	1984-05-27	0.25	NA	NA	1550.0
L	Paul Lake	1984	148	1984-05-27	0.50	NA	NA	1550.0
L	Paul Lake	1984	148	1984-05-27	0.75	NA	NA	1550.0
L	Paul Lake	1984	148	1984-05-27	1.00	14.5	9.5	1750.0
L	Paul Lake	1984	148	1984-05-27	1.50	NA	NA	1550.0
L	Paul Lake	1984	148	1984-05-27	2.00	14.2	9.5	1750.0
L	Paul Lake	1984	148	1984-05-27	3.00	11.0	9.5	1750.0
L	Paul Lake	1984	148	1984-05-27	4.00	7.0	9.5	1750.0
L	Paul Lake	1984	148	1984-05-27	5.00	6.1	9.5	1750.0

lakename	sampledate	depth	temperature_C	temperature_F
Paul Lake	1984-05-27	0.00	14.5	58.10
Paul Lake	1984-05-27	0.25	NA	NA
Paul Lake	1984-05-27	0.50	NA	NA
Paul Lake	1984-05-27	0.75	NA	NA
Paul Lake	1984-05-27	1.00	14.5	58.10
Paul Lake	1984-05-27	1.50	NA	NA
Paul Lake	1984-05-27	2.00	14.2	57.56
Paul Lake	1984-05-27	3.00	11.0	51.80
Paul Lake	1984-05-27	4.00	7.0	44.60
Paul Lake	1984-05-27	5.00	6.1	42.98

Q&A: dplyr

Filter	Arrange	Select	Mutate	Pipe	Lubridate
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Subset/rearrange columns...

lakeid	lakename	year4	daynum	sampledate	depth	temperature_C	dissolvedOxygen
L	Paul Lake	1984	148	1984-05-27	0.00	14.5	9.5
L	Paul Lake	1984	148	1984-05-27	0.25	NA	NA
L	Paul Lake	1984	148	1984-05-27	0.50	NA	NA
L	Paul Lake	1984	148	1984-05-27	0.75	NA	NA
L	Paul Lake	1984	148	1984-05-27	1.00	14.5	8.8
L	Paul Lake	1984	148	1984-05-27	1.50	NA	NA
L	Paul Lake	1984	148	1984-05-27	2.00	14.2	8.6
L	Paul Lake	1984	148	1984-05-27	3.00	11.0	11.5
L	Paul Lake	1984	148	1984-05-27	4.00	7.0	11.9
L	Paul Lake	1984	148	1984-05-27	5.00	6.1	2.5



year4	lakeid	depth
1984	L	0.00
1984	L	0.25
1984	L	0.50
1984	L	0.75
1984	L	1.00
1984	L	1.50
1984	L	2.00
1984	L	3.00
1984	L	4.00
1984	L	5.00

Q&A: dplyr

Filter	Arrange	Select	Mutate	Pipe	Lubridate
--------	---------	--------	--------	------	-----------

Calculate a column of new values from existing ones

lakeid	lakename	year4	daynum	sampldate	depth	temperature_C	dissolvedOxygen	T_x_DO
L	Paul Lake	1984	148	1984-05-27	0.00	14.5	9.5	137.75
L	Paul Lake	1984	148	1984-05-27	0.25	NA	NA	NA
L	Paul Lake	1984	148	1984-05-27	0.50	NA	NA	NA
L	Paul Lake	1984	148	1984-05-27	0.75	NA	NA	NA
L	Paul Lake	1984	148	1984-05-27	1.00	14.5	8.8	127.60
L	Paul Lake	1984	148	1984-05-27	1.50	NA	NA	NA
L	Paul Lake	1984	148	1984-05-27	2.00	14.2	8.6	122.12
L	Paul Lake	1984	148	1984-05-27	3.00	11.0	11.5	126.50
L	Paul Lake	1984	148	1984-05-27	4.00	7.0	11.9	83.30
L	Paul Lake	1984	148	1984-05-27	5.00	6.1	2.5	15.25

Q&A: dplyr

Filter	Arrange	Select	Mutate	Pipe	Lubridate
--------	---------	--------	--------	------	-----------

Subset rows based on a criteria

lakeid	lakename	year4	daynum	sampldate	depth	temperature_C	dissolvedOxygen
L	Paul Lake	1984	148	1984-05-27	0.00	14.5	9.5
L	Paul Lake	1984	148	1984-05-27	0.25	NA	NA
L	Paul Lake	1984	148	1984-05-27	0.50	NA	NA
L	Paul Lake	1984	148	1984-05-27	0.75	NA	NA
L	Paul Lake	1984	148	1984-05-27	1.00	14.5	8.8
L	Paul Lake	1984	148	1984-05-27	1.50	NA	NA
L	Paul Lake	1984	148	1984-05-27	2.00	14.2	8.6
lakeid	lakename	year4	daynum	sampldate	depth	temperature_C	dissolvedOxygen
L	Paul Lake	1984	148	1984-05-27	0	14.5	9.5
R	Peter Lake	1984	149	1984-05-28	0	14.8	9.2
T	Tuesday Lake	1984	150	1984-05-29	0	15.0	9.5
L	Paul Lake	1984	155	1984-06-03	0	18.8	8.0
R	Peter Lake	1984	156	1984-06-04	0	18.8	9.0
T	Tuesday Lake	1984	157	1984-06-05	0	21.0	8.4
L	Paul Lake	1984	162	1984-06-10	0	19.6	8.5
R	Peter Lake	1984	163	1984-06-11	0	19.8	8.9
T	Tuesday Lake	1984	164	1984-06-12	0	20.4	8.9
L	Paul Lake	1984	169	1984-06-17	0	21.0	7.3

Q&A: dplyr

Filter	Arrange	Select	Mutate	Pipe	Lubridate
--------	---------	--------	--------	------	-----------

Sort rows based on values in one or more columns...

lakeid	lakename	year4	daynum	sampldate	depth	temperature_C	dissolvedOxygen
L	Paul Lake	1984	148	1984-05-27	0.00	14.5	9.5
L	Paul Lake	1984	148	1984-05-27	0.25	NA	NA
L	Paul Lake	1984	148	1984-05-27	0.50	NA	NA
L	Paul Lake	1984	148	1984-05-27	0.75	NA	NA
L	Paul Lake	1984	148	1984-05-27	1.00	14.5	8.8
L	Paul Lake	1984	148	1984-05-27	1.50	NA	NA
lakeid	lakename	year4	daynum	sampldate	depth	temperature_C	dissolvedOxygen
L	Paul Lake	1984					
L	Paul Lake	1984					
L	Paul Lake	1984					
L	Paul Lake	1984					
T	Tuesday Lake	1987	195	1987-07-14	12.0	0.3	0.1
T	Tuesday Lake	1988	195	1988-07-13	12.0	0.3	0.1
R	Peter Lake	1989	157	1989-06-06	12.0	0.7	4.3
R	Peter Lake	2000	145	2000-05-24	12.0	1.1	4.4
C	Central Long Lake	1994	217	1994-08-05	3.5	1.3	NA
R	Peter Lake	1989	157	1989-06-06	10.0	1.4	4.6
R	Peter Lake	2000	145	2000-05-24	11.0	1.6	4.4
T	Tuesday Lake	1985	177	1985-06-26	7.0	2.8	NA
T	Tuesday Lake	1985	177	1985-06-26	8.0	2.8	NA
T	Tuesday Lake	1985	177	1985-06-26	10.0	2.8	NA

Tips: Handy shortcuts

Key Combination	Result
<Alt> + -	<-
<Shift> + <Ctrl> + M	%>%
<Shift> + <Alt> + I	<i>New code chunk</i>

Tips: Debugging piped commands

Part 2 - “split-apply-combine” code chunk

```
NTL.PeterPaul.summaries <-  
  NTL.phys.nutrient.data.PeterPaul %>%  
  filter(depth == 0) %>%  
  group_by(lakename, month) %>%  
  filter(  
    !is.na(temperature_C) &  
    !is.na(tn_ug) &  
    !is.na(tp_ug)  
  ) %>%  
  summarise(  
    meantemp = mean(temperature_C),  
    sdtemp = sd(temperature_C),  
    meanTN = mean(tn_ug),  
    sdTN = sd(tn_ug),  
    meanTP = mean(tp_ug),  
    sdTP = sd(tp_ug)  
  )
```

NTL.PeterPaul.summaries 1069 obs. of 15 variables

NTL.PeterPaul.summaries 181 obs. of 15 variables

NTL.PeterPaul.summaries 10 obs. of 8 variables

- Select a subset of the code and run; check the output

M4.2 – Data Wrangling II

Q&A on Data *Pipeline*, transform, grouping

- **Data pipeline:**

- *Session set-up / Import & Explore / Wrangle*

- **More wrangling**

- Gather (pivot-longer) & Spread (pivot-wider)
 - Joining datasets
 - Grouping & summarizing data

M4.3 – Data Wrangling III (lab)

1. Import and wrangle

- The data:

- <https://lter.limnology.wisc.edu/about/overview>

- ▣ Nutrient data, Physical data

- ▣ Peter and Paul Lakes ([Link](#))



- Import, explore, wrangle

- ▣ Subset for Peter and Paul Lakes

- ▣ Fix dates

- ▣ Filtering (on multiple values with `%in%`)

Exercise 1 & 2: Filtering

- Filter “NTL.phys.data” for the year 1999
 - ▣ Should get 1898 rows
- Filter for Tuesday Lake records from 1990 thru 1999
 - ▣ Should get 1971 rows

Exercise: Add pipes

```
103 ▾ ```{r Exercise 3 & 4}
104 #Example using pipes to wrangle data:
105 #Add pipes in the correct place below
106 NTL.phys.data.processed <-
107   NTL.phys.data
108   filter(lakename == "Paul Lake" | lakename == "Peter Lake")
109   select(lakename, sampledate:temperature_C)
110   mutate(temperature_F = (temperature_C*9/5) + 32)
```

Exercise 3: Pipes

- Using pipes: Filter “NTL.phys.data” for:
 - ▣ Tuesday Lake
 - ▣ from 1990 through 1999
 - ▣ only for records collected in July

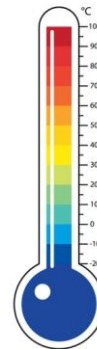
* Tip: you may want to create a new column of just the month

Should get 537 rows

Exercise 4: Pipes

- Using the data from part 3, pipes, and the `summarize()` function, find the mean surface temperature...
 1. Need to subset for surface records... (25 obs)
 2. Good practice to eliminate NAs (still 25 obs)
 3. `summarize()` to compute means on a column (1 obs)

22.46 °C



2. Reshape the nutrient data



04_Part2_DataWrangling.Rmd

```
90 # Gather nutrient data into one column using pivot_longer
91 NTL.nutrient.data.PeterPaul.gathered <-
  pivot_longer(NTL.nutrient.data.PeterPaul.subset, tn_ug:po4, names_to =
    "nutrient", values_to = "concentration")
```

	lakename	year4	daynum	month	sampledate	depth	tn_ug	tp_ug	nh34	no23	po4
1	Paul Lake	1991	140	5	1991-05-20	0.00	538	25	NA	NA	NA
2	Paul Lake	1991	140	5	1991-05-20	0.85	285	14	NA	NA	NA
3	Paul Lake	1991	140	5	1991-05-20	1.75	399	14	NA	NA	NA
4	Paul Lake	1991	140	5	1991-05-20	3.00	453	14	NA	NA	NA
5	Paul Lake	1991	140	5	1991-05-20	4.00	363	13	NA	NA	NA
6	Paul Lake	1991	140	5	1991-05-20	6.00	583	37	NA	NA	NA

cols

names_to values_to

	lakename	year4	daynum	month	sampledate	depth	nutrient	concentration
1	Paul Lake	1991	140	5	1991-05-20	0.00	tn_ug	538.000
2	Paul Lake	1991	140	5	1991-05-20	0.00	tp_ug	25.000
3	Paul Lake	1991	140	5	1991-05-20	0.00	nh34	NA
4	Paul Lake	1991	140	5	1991-05-20	0.00	no23	NA
5	Paul Lake	1991	140	5	1991-05-20	0.00	po4	NA
6	Paul Lake	1991	140	5	1991-05-20	0.85	tn_ug	285.000
7	Paul Lake	1991	140	5	1991-05-20	0.85	tp_ug	14.000

Exercise 5: *pivot_longer()*

cols

lakeid	lakename	year4	daynum	sampledate	depth	temperature_C	dissolvedOxygen	irradianceWater	irradianceDeck	comments
L	Paul Lake	1984	148	1984-05-27	0.00	14.5	9.5	1750.0	1620	NA
L	Paul Lake	1984	148	1984-05-27	0.25	NA	NA	1550.0	1620	NA
L	Paul Lake	1984	148	1984-05-27	0.50	NA	NA	1150.0	1620	NA
L	Paul Lake	1984	148	1984-05-27	0.75	NA	NA	975.0	1620	NA
L	Paul Lake	1984	148	1984-05-27	1.00	14.5	8.8	870.0	1620	NA
L	Paul Lake	1984	148	1984-05-27	1.50	NA	NA	610.0	1620	NA
L	Paul Lake	1984	148	1984-05-27	2.00	14.2	8.6	420.0	1620	NA
L	Paul Lake	1984	148	1984-05-27	3.00	11.0	11.5	220.0	1620	NA

names_to

values_to

lakeid	lakename	year4	daynum	sampledate	depth	temperature_C	dissolvedOxygen	comments	irradiance_type	irradiance
L	Paul Lake	1984	148	1984-05-27	0.00	14.5	9.5	NA	irradianceWater	1750.0
L	Paul Lake	1984	148	1984-05-27	0.00	14.5	9.5	NA	irradianceDeck	1620.0
L	Paul Lake	1984	148	1984-05-27	0.25	NA	NA	NA	irradianceWater	1550.0
L	Paul Lake	1984	148	1984-05-27	0.25	NA	NA	NA	irradianceDeck	1620.0
L	Paul Lake	1984	148	1984-05-27	0.50	NA	NA	NA	irradianceWater	1150.0
L	Paul Lake	1984	148	1984-05-27	0.50	NA	NA	NA	irradianceDeck	1620.0
L	Paul Lake	1984	148	1984-05-27	0.75	NA	NA	NA	irradianceWater	975.0
L	Paul Lake	1984	148	1984-05-27	0.75	NA	NA	NA	irradianceDeck	1620.0

Exercise 5: *pivot_wider()*

names_from values_from

lakeid	lakename	year4	daynum	sampledate	depth	temperature_C	dissolvedOxygen	irradianceWater	irradianceDeck
L	Paul Lake	1984	148	1984-05-27	0.00	14.5	9.5	1750.0	1620
L	Paul Lake	1984	148	1984-05-27	0.25	NA	NA	1550.0	1620
L	Paul Lake	1984	148	1984-05-27	0.50	NA	NA	1150.0	1620
L	Paul Lake	1984	148	1984-05-27	0.75	NA	NA	975.0	1620
L	Paul Lake	1984	148	1984-05-27	1.00	14.5	8.8	870.0	1620
L	Paul Lake	1984	148	1984-05-27	1.50	NA	NA	610.0	1620
L	Paul Lake	1984	148	1984-05-27	2.00	14.2	8.6	420.0	1620
L	Paul Lake	1984	148	1984-05-27	3.00	11.0	11.5	220.0	1620
L	Paul Lake	1984	148	1984-05-27	4.00	7.0	11.9	100.0	1620

id_cols
(or select columns
before pivot)

sampledate	0	0.25	0.5	0.75	1	1.5	2	3	4	5	6	7	8	9	10
1984-05-27	14.5	NA	NA	NA	14.5	NA	14.2	11.0	7.0	6.1	5.5	5.0	4.5	4.5	4.5
1984-05-28	14.8	NA	NA	NA	14.8	NA	14.8	12.3	8.2	7.0	5.9	4.5	4.0	4.0	3.9
1984-05-29	15.0	NA	NA	NA	14.5	14.0	10.5	6.8	5.3	5.0	4.5	4.0	4.0	3.9	3.9
1984-06-03	18.8	NA	18.8	NA	18.7	18.3	17.0	13.0	9.0	6.7	5.8	5.0	4.8	NA	4.7
1984-06-04	18.8	NA	18.8	NA	18.8	18.5	18.0	14.7	10.1	7.5	6.0	5.0	4.4	NA	4.0
1984-06-05	21.0	NA	21.0	NA	20.2	16.9	12.4	7.1	5.7	5.0	4.6	NA	4.0	NA	3.9
1984-06-10	19.6	NA	19.6	NA	19.6	19.4	19.2	14.4	10.0	7.3	6.2	5.2	4.9	4.8	4.8
1984-06-11	19.8	NA	19.9	NA	19.9	20.0	19.9	15.9	11.3	8.0	5.9	4.9	4.6	4.1	4.0
1984-06-12	20.4	NA	20.4	NA	20.1	18.6	14.4	8.0	5.9	5.0	4.7	4.2	4.0	NA	4.0
1984-06-17	21.0	NA	21.0	NA	20.8	20.5	20.2	15.7	10.7	7.8	6.5	5.4	5.0	5.0	4.9
1984-06-18	20.7	NA	20.8	NA	20.8	20.8	20.5	17.9	12.5	8.7	6.4	5.2	4.7	NA	4.1